

CU Hyperloop

Electrical & Software Team Report

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August 2024 – September 2025

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Project Overview

Parameter	Detail
Organization	University of Colorado Boulder — CU Hyperloop
Role	Electrical & Software Team Member
Duration	August 2024 – September 2025
Competition	Not a Boring Competition (hosted by The Boring Company)

Table 1: CU Hyperloop project summary

The objective of the CU Hyperloop electrical and software team was to develop the electrical systems and embedded firmware for a student-built Tunnel Boring Machine (TBM). The system was designed to compete in The Boring Company’s annual *Not a Boring Competition*, aiming to excavate a 100 ft tunnel with stable structural support.

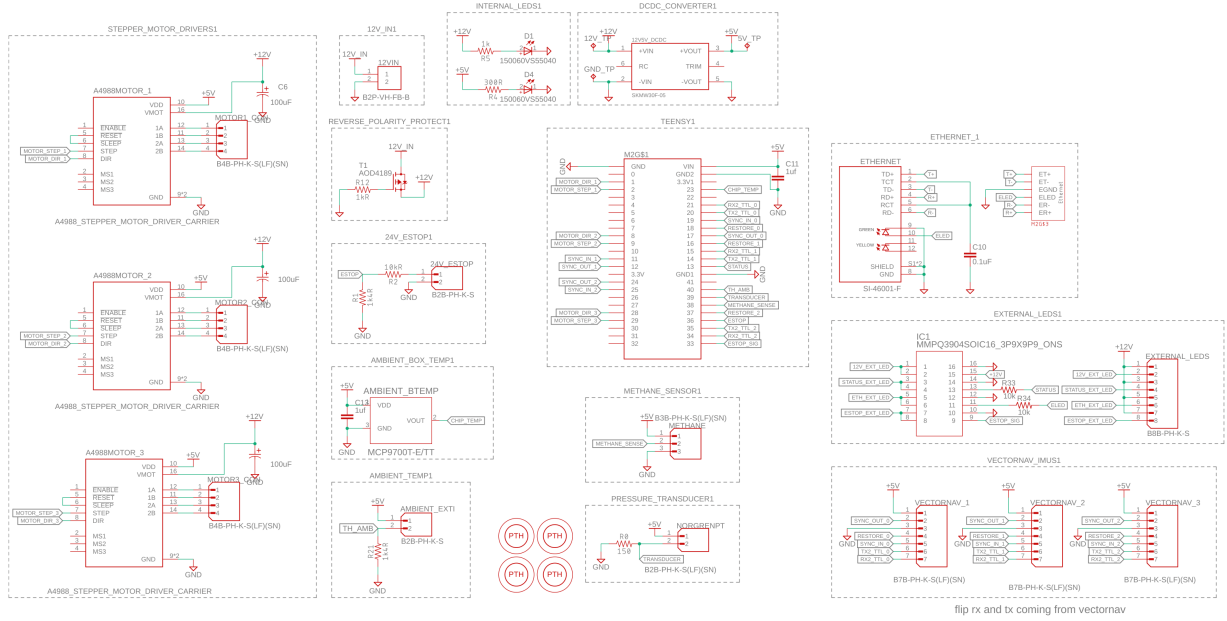
PCB Design

Fluids Board

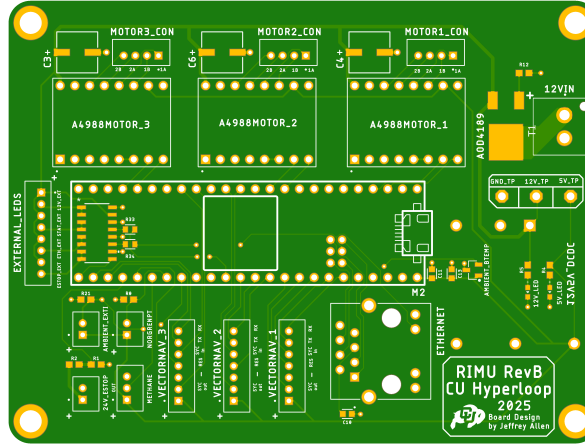
The Fluids Board was designed to control six distinct pumps serving two critical subsystems: Ground Control Station (GCS) cooling and internal TBM system cooling.

Channel(s)	Load	Topology / Notes
CH 1–2	GCS Pumps (2×)	Buck converter with internal feedback loop
CH 3–6	Internal Cooling (4×)	Direct drive
<i>Microcontroller:</i> Teensy 3.1		

Table 2: Fluids board channel allocation



(a) Navigation Board Schematic Revision B



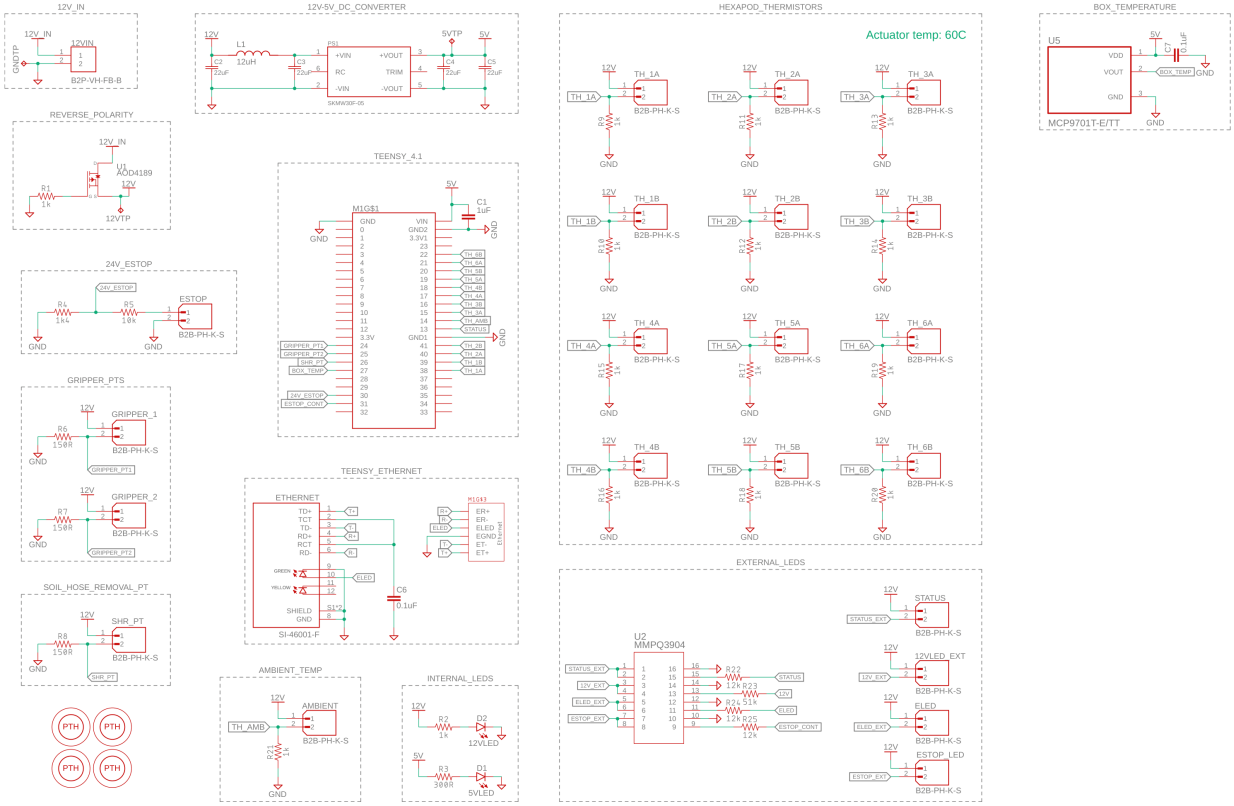
(b) Navigation Board Revision B PCB Layout

Figure 2: Revision B of the Navigation Board. Provided positioning and drift data utilizing VectorNav-100 IMUs, acquiring and transmitting telemetry to the overall system GUI via Ethernet.

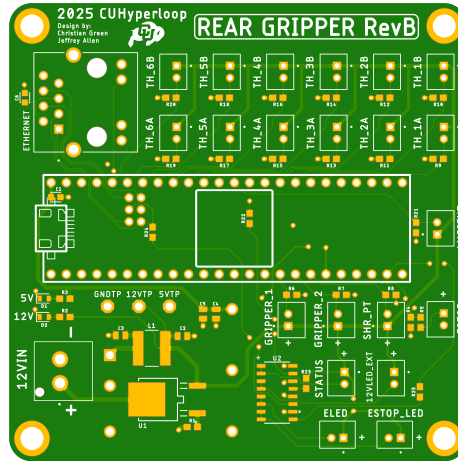
Rear Gripper Board

The Rear Gripper Board is a JST-connector-dense PCB engineered to capture and transmit environmental data to the master control system. It processes telemetry regarding temperature, methane presence, and pressure specifically for the 6-linear-actuator pushing system. I was the sole designer for Revision A, and I handled the circuit design for Revision B, with

Christian Green handling the board layout.



(a) Rear Gripper Board Schematic Revision B



(b) Rear Gripper Board Revision B PCB Layout

Figure 3: Revision B of the Rear Gripper Board. Designed specifically for localized data acquisition at the rear of the Tunnel Boring Machine.

Relevant Skills Demonstrated

Category	Skills / Tools
PCB Engineering	Altium Designer, schematic capture, layout, multi-revision bring-up
Power Converter Topologies	Buck converter with internal feedback (Fluids Board)
Embedded Firmware	C/C++, Teensy 3.1 (ARM Cortex-M4), Arduino framework
Communication Protocols	EtherCAT, Ethernet, SPI, I ² C, UART
Sensor Integration	VectorNav-100 IMU, methane sensors, temperature sensors, pressure sensors
Mechanical CAD	Enclosure dimensioning, connector placement for TBM integration
Technical Documentation	BOM structuring, Revision control, design rationale documentation

Table 3: Skills demonstrated during CU Hyperloop